

Task 1: List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Scenario	How the multimeter helps
Laptop won't power on	Set to DC voltage mode and set the +probe at the inner round barrel connector and -ve (grd) probes at the outer round barrel connector, check actually delivering voltage is matching the rated voltage of the charger.
Smartphone Screen Backlight Failure	Set to DC voltage mode, a multimeter is used to measure the output of the backlight boost circuit at the display connector pins while the phone is powered on, confirming whether the motherboard is successfully generating the high voltage (typically 15V to 40V) required to illuminate the screen's LEDs.
Laptop battery charging fault	With the battery removed and the AC adapter plugged in, power on the laptop., touch the red probe to the positive pin and black probe to a ground/negative pin to check battery charging pins output voltages.

Task 2 :Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.Hint: Focus on the codes shown and how they relate to hardware issues.

A POST (Power-On Self-Test) diagnostic card plugs into a PCI, ISA, or laptop miniPCI/miniPCIe slot and reveals the BIOS's internal progress as it boots.

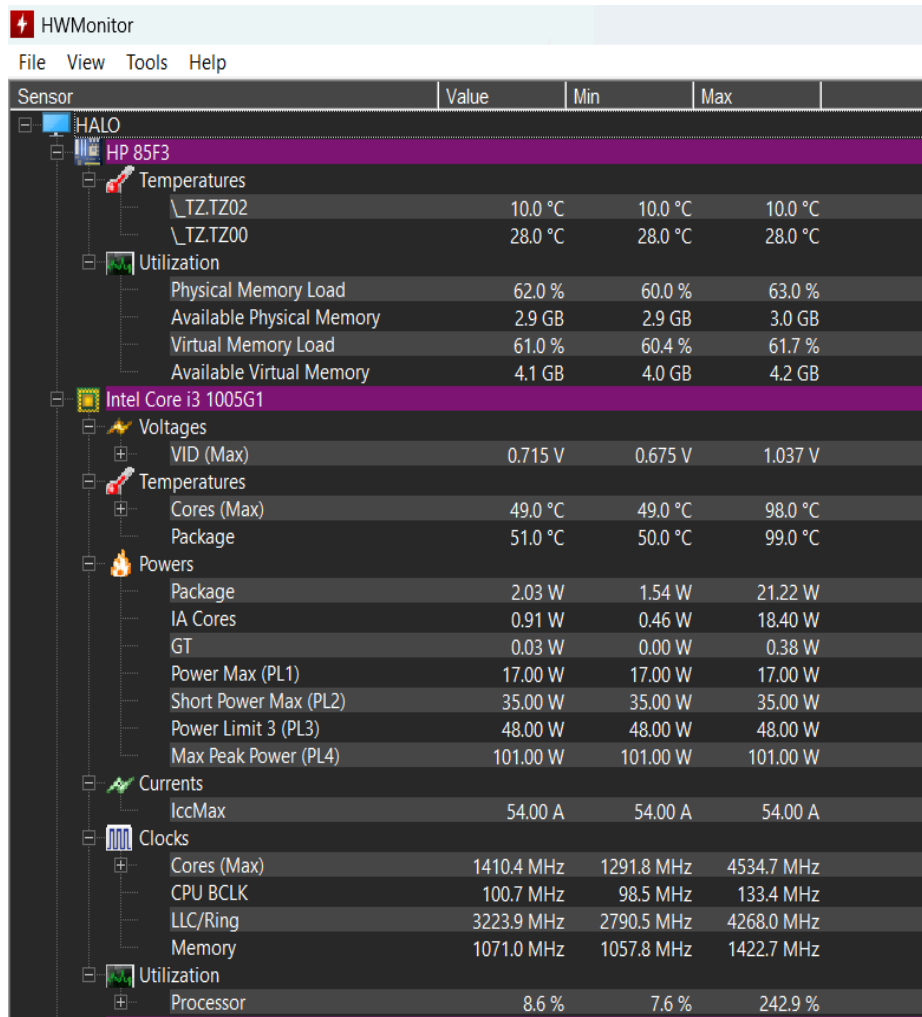
LED diagnosis: Alongside the hex code, the card's LEDs (CLK and RST being the most critical) confirm basic board health even when the hex display shows nothing at all. The **RST (reset) LED** should flash briefly once or twice right after power-on as the board completes its reset cycle.

Beep code diagnosis: If a board fails before even reaching video initialization, the BIOS communicates through a series of beeps instead. For example, on AMI BIOS, **5 short beeps** specifically means *"Processor error — the system CPU has failed,"* immediately narrowing the fault to the CPU rather than RAM or video.

Hex code diagnosis: The card shows a two-digit hex code corresponding to the exact BIOS routine currently running. The video showed a code **2C** means *"RAM failure on*

address line" — a card stuck at this code points directly to a bad or loose memory module.

Task 3: Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.



HWMonitor				
File View Tools Help				
Sensor	Value	Min	Max	
TOSHIBA MQ04ABF100 (70IDPESUT)				
Temperatures				
Assembly	37.0 °C	37.0 °C	38.0 °C	
Counters				
Power-On-Hours	6172 hrs	6172 hrs	6172 hrs	
Power Cycle Count	3959	3959	3959	
Utilization				
Used Space (d:)	17.4 %	17.4 %	17.4 %	
Available Capacity (d:)	204.8 GB	204.8 GB	204.8 GB	
Used Space (e:)	10.5 %	10.5 %	10.5 %	
Available Capacity (e:)	227.2 GB	227.2 GB	227.2 GB	
Used Space (f:)	31.6 %	31.6 %	31.6 %	
Available Capacity (f:)	293.8 GB	293.8 GB	293.8 GB	
Activity	0.0 %	0.0 %	25.5 %	
Speed				
Read Rate	0.00 MB/s	0.00 MB/s	0.02 MB/s	
Write Rate	0.00 MB/s	0.00 MB/s	0.00 MB/s	
POWER X NE1000 (DWP2309256K3377)				
Temperatures				
Assembly	54.0 °C	54.0 °C	54.0 °C	
Counters				
Power-On-Hours	2850 hrs	2850 hrs	2850 hrs	
Power Cycle Count	2448	2448	2448	
Health Status	97 %	97 %	97 %	
Utilization				
Used Space (c:)	57.9 %	57.9 %	57.9 %	
Available Capacity (c:)	100.0 GB	99.9 GB	100.0 GB	
Activity	15.0 %	0.0 %	84.6 %	
Speed				
Read Rate	66.13 MB/s	0.00 MB/s	66.13 MB/s	
Write Rate	0.78 MB/s	0.00 MB/s	10.56 MB/s	
Intel(R) UHD Graphics				
Utilization				
GPU	0.9 %	0.0 %	33.2 %	
Video Engine	0.0 %	0.0 %	0.0 %	
3D	0.9 %	0.0 %	33.2 %	
Copy	0.0 %	0.0 %	0.0 %	
Compute	0.0 %	0.0 %	0.0 %	
Dedicated Memory	0.0 MB	0.0 MB	0.0 MB	
Shared Memory	317.5 MB	300.3 MB	334.0 MB	
PABAS0241231				
Voltages				
Current Voltage	11.900 V	11.806 V	11.900 V	
Powers				
Charge Rate	29.32 W	28.10 W	29.45 W	
Capacities				
Designed Capacity	36000 mWh	36000 mWh	36000 mWh	
Full Charge Capacity	36000 mWh	36000 mWh	36000 mWh	
Current Capacity	15060 mWh	13860 mWh	15060 mWh	
Levels				
Wear Level	0.0 %	0.0 %	0.0 %	
Charge Level	41.8 %	38.5 %	41.8 %	
VirtualBox Host-Only Ethernet Adapter				
Speed				
Download Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps	
Upload Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps	

Task 4: Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Voltage, continuity, resistance at specific points/pins	A laptop DC jack shows 0V output, confirming a faulty charger rather than a dead motherboard
POST card	BIOS Power-On Self-Test progress via hex codes, for machines that won't boot to a display	Code freezes at the memory-test stage, pointing to a bad or improperly seated RAM module
Software diagnostics (HWMonitor/CrystalDiskInfo)	Live system health: temperatures, voltages, fan speeds, drive SMART status	CPU idling at 90°C flags failing thermal paste/fan, or a drive's SMART status shows "Caution," warning of imminent failure